

Zaryan Syed

Milton, ON | (647) 917-0212 | zaryan@uoguelph.ca | linkedin.com/in/zaryan-syed

TECHNICAL SKILLS

Programming & HDL: C, C++, VHDL, Java, JavaScript, MATLAB

Digital Hardware: FPGA Design, Digital Logic, CPU Datapaths, ALUs, Register Files, FSMs, Memory Systems, UART

Verification & Testing: VHDL Testbenches, Functional Verification, Simulation, Waveform Analysis, Timing Analysis, Hardware Debugging

Tools & Systems: Vivado, Nexys A7 FPGA, Linux, Git, GCC, Make, Docker, VS Code, Arduino

EDUCATION

University of Guelph

Bachelor of Engineering, Computer Engineering Co-op

Guelph, ON

Sept. 2024 – Apr. 2029

- **Relevant Coursework:** Computer Organization, Digital Systems, Embedded Systems, Microcomputer Interfacing, Signal Processing, Electronic Devices, Operating Systems, Large-Scale Software Architecture

HARDWARE & SOFTWARE PROJECTS

FPGA CPU & Hardware-Software Interface | VHDL, UART, Vivado, Nexys A7 FPGA

Sept. 2025 – Present

- Designed and integrated a 16-bit single-cycle CPU from 14 VHDL modules, including ALU, register file, memory, multiplexers, control logic, and program-counter logic.
- Achieved a 100% pass rate across 32 functional test cases using VHDL testbenches and waveform analysis to verify CPU datapath and control behaviour.
- Integrated a 9600-baud UART interface for host-to-FPGA communication and processor-state readback, enabling external testing and debugging.

Round Robin CPU Scheduler | C, POSIX Threads, Linux

Jun. 2026 - Aug. 2026

- Built a deterministic Round Robin scheduler in C modelling 5 process states with configurable time quantum, blocking, and ready-queue management.
- Validated scheduler behaviour across 3 workloads and quantum values 1–5, handling arrivals, blocking, wakeups, quantum expiry, requeueing, and idle CPU periods.
- Calculated 4 scheduling metrics per task, with tested workloads producing up to 14 CPU dispatches.

University Student Management System | Java, JavaFX, OOP

Jan. 2025 – Apr. 2025

- Developed a JavaFX application with 10+ screens and 7 core features supporting student, faculty, and administrator workflows.
- Supported student management across 6 university courses and successfully demonstrated authentication, dashboards, faculty management, and session workflows.

GPYOU – AI GPU Recommendation Platform | JavaScript, Gemini API

Apr. 2025 – May 2025

- Built a functional GPU recommendation platform in a 5-person team within 48 hours, matching users to 12 GPUs using 4 preference inputs.
- Structured GPU specifications for Gemini-based matching and helped deliver a 3-screen application that won Best UI among 50 competing projects.

EXPERIENCE

Tim Hortons

Team Member

Milton, ON

Nov. 2022 – Present

- Collaborate in teams of 10–12 employees to prioritize tasks and resolve operational issues in a high-volume environment.
- Support customer service, order preparation, cash handling, inventory, and sanitation across multiple workstations.

ASW Works

Construction Assistant

Milton, ON

Apr. 2025 – Sept. 2025

- Supported crews of up to 4 across 3–5 residential renovation projects, adapting installations to changing site requirements.
- Performed precision measurement, framing, drywall, flooring, and material preparation using 10+ hand and power tools.